

Routing-Method Effects on Distance, Time, Fuel, and Emissions in Europe–Asia Trade: A Comparison of the Suez, Cape, and Northern Sea Route Corridors

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Abstract

Growing interest in decarbonization and Arctic accessibility has renewed attention on Europe–Asia shipping corridors. The Northern Sea Route (NSR) is often portrayed as a 30–40% shortcut relative to Suez, with savings propagated to time, fuel, and CO₂. The effect of enforcing sea-only feasibility on these baselines, and its downstream impact on time, fuel, and CO₂, remains under-examined.

We compare great-circle baselines with sea-only routes computed via A-star search (A*) on a 0.5° grid between Northern Europe and Northeast Asia across the Suez, Cape of Good Hope, and NSR corridors under three waypoint philosophies. Distances are mapped to voyage time using corridor-typical speeds and to fuel/CO₂ using main- and auxiliary-engine accounting.

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Sea-only routing preserves the ranking $\text{NSR} < \text{Suez} < \text{Cape}$ but compresses NSR’s advantage once realistic speeds are applied. NSR remains shortest (about 8–10k nm versus 11–12k nm for Suez), yet typical durations differ modestly and fuel/CO₂ savings over Suez are small and variant-dependent. Equal-speed tests restore geometric ordering, and endpoint sensitivity shows larger NSR gains for more northern East Asian ports.

The framework provides a reproducible, corridor-agnostic benchmark for later integration of sea ice, weather, regulatory overlays, and AIS data in dynamic Arctic voyage planning.

Keywords: sea-only routing, A* shortest-path search, great-circle baseline, corridor benchmarking, Suez Canal, Cape of Good Hope, Northern Sea Route (NSR), fuel consumption, CO₂ emissions, Arctic shipping

Nomenclature

A*	A-star Pathfinding Algorithm
AE	Auxiliary Engine
AIS	Automatic Identification System
CO ₂	Carbon Dioxide
DWT	Deadweight Tonnage
GC	Great Circle
GHG	Greenhouse Gas
IMO	International Maritime Organization
ME	Main Engine
NSR	Northern Sea Route
PM	Particulate Matter

SUEZ	Suez Canal
TSS	Traffic Separation Schemes

1. Introduction

Growing attention to decarbonization and Arctic accessibility has renewed interest in alternative Europe–Asia shipping corridors. The Northern Sea Route (NSR) is frequently presented as a 30–40% shorter alternative to Suez based on great-circle (GC) geometry (Liu and Kronbak, 2010; Furuichi and Otsuka, 2015; Schøyen and Bråthen, 2011). Recent corridor assessments continue to report NSR advantages primarily from geometric or stylized baselines before adding cost or feasibility layers (Li et al., 2023; Zeng et al., 2020). Some studies propagate these geometric savings to time, fuel, or CO₂ without first demonstrating sea-only navigability (International Association of Ports and Harbors (IAPH), Port Planning and Development Committee, 2013; Wan et al., 2018), whereas others explicitly model Arctic feasibility and operating speeds, often jointly with regulatory or emissions-control layers (Smith and Stephenson, 2013; Melia et al., 2016; Kavirathna et al., 2023). Because corridor baselines are increasingly used to parameterize network, climate, and policy models, small routing-representation biases can scale into large system-level conclusions (Poo et al., 2024). What remains under-examined is how enforcing sea-only feasibility reshapes corridor baselines and their downstream translation to time, fuel, and CO₂ under corridor-specific speeds.

We compare geometric GC baselines (direct GC and GC via corridor macro-waypoints) with physically feasible sea-only routes computed on a coastline-masked 0.5° A* graph. The framework is applied to Rotterdam–Yokohama, a representative Europe–Asia pair spanning the Suez, Cape of Good Hope, and NSR corridors. Each corridor is represented by three waypoint variants to capture distinct routing philosophies: (i) *service-guided* (hub/chokepoint oriented), (ii) *bluewater-oriented* (open-ocean favouring), and (iii) *channel/coast-guided* (lane/coast following). This design magnifies routing-method effects over an intercontinental distance while remaining comparable to prior corridor studies (Li et al., 2023; Kavirathna et al., 2023; Meza et al., 2023). Distances are mapped to voyage time using corridor-typical service speeds and to fuel/CO₂ using explicit main- and auxiliary-

engine accounting. To isolate routing effects, we exclude sea ice, weather, emission control areas (ECAs), bathymetry and draft limits, and AIS-based tracking; these layers are reserved for subsequent constrained analyses so that environmental and regulatory factors do not confound the methodological comparison.

Our analysis yields three headline insights. First, enforcing sea-only feasibility preserves the distance ranking ($\text{NSR} < \text{Suez} < \text{Cape}$): NSR remains shortest once routes are rendered navigable. Second, distance does not translate linearly to time or fuel—corridor-specific speeds and practices such as slow steaming can compress, or even eliminate, NSR’s apparent operational advantage over Suez (Notteboom and Vernimmen, 2009; Maloni et al., 2013). Third, although longest, the Cape corridor remains a robust year-round fallback independent of canals and Arctic seasonality, consistent with the recent diversion of container services around Africa during the Red Sea crisis (?).

We focus on Europe–Asia trade because it is a major deep-sea liner market and a key arena for debates on the competitiveness of alternative long-haul corridors. Rotterdam–Yokohama spans the Suez, Cape of Good Hope, and NSR corridors and is widely used as a representative origin–destination pair in the NSR–Suez literature. Recent route-comparison studies likewise use Rotterdam–Asia pairs to quantify Arctic versus Suez corridor differences (Meza et al., 2023).

Macro-waypoints were placed at natural gateway regions (canal entrances, straits, capes). Each successive pair was verified to admit a valid sea-only path in `searoute`; cases with land intersections or graph failures were iteratively adjusted and then checked visually on interactive charts.

From the reviewed literature, three methodological gaps emerge:

- (i) **Method isolation is rare.** Comparative studies often remain geometric (GC) or embed routing within complex environmental or economic models, obscuring how routing representation itself biases distance or derived efficiency metrics (Liu and Kronbak, 2010; Schøyen and Bråthen, 2011; Smith and Stephenson, 2013; Melia et al., 2016).
- (ii) **Propagation into emissions is underexplored.** Few works trace routing assumptions through to fuel and CO₂ accounting with explicit treatments of main- and auxiliary-engine loads (Schröder et al., 2017; Chen et al., 2021; Karamperidis and coauthors, 2022; Johansson et al., 2022; Nguyen and coauthors, 2023).

- (iii) **Reproducibility is uneven.** Many studies provide limited access to code, parameters, or scenario definitions, hindering cross-study benchmarking and robustness checks.

The primary objective is to quantify how routing representation (GC versus sea-only A* on a 0.5° water mask) alters corridor distances and their translation to voyage time, fuel, and CO₂ for Rotterdam–Yokohama across Suez, Cape of Good Hope, and NSR. Secondary objectives are to contrast corridor-typical versus equal-speed baselines, test endpoint sensitivity (for example, Busan versus Yokohama).

Contributions. This study provides:

- (a) a corridor-agnostic, reproducible sea-only benchmark for Europe–Asia routing across Suez, Cape of Good Hope, and NSR;
- (b) quantified GC→sea-only deltas under three waypoint philosophies;
- (c) transparent propagation to time and fuel/CO₂ using explicit main- and auxiliary-engine accounting; and
- (d) robustness and endpoint sensitivity analyses identifying when NSR’s geometric advantage does—and does not—translate into operational savings.

2. Data and Methods

2.1. Study scope and corridor design

The analysis targets deep-sea services between Northern Europe and Northeast Asia, where the choice of long-haul corridor has the largest implications for distance, time, fuel, and emissions. We select Rotterdam (51.95°N , 4.14°E) and Yokohama (35.45°N , 139.65°E) as the baseline origin–destination pair because it:

- lies on a major Europe–Asia trade lane,
- is widely used in comparative corridor studies, and
- can be served via Suez, Cape of Good Hope, or the Northern Sea Route (NSR).

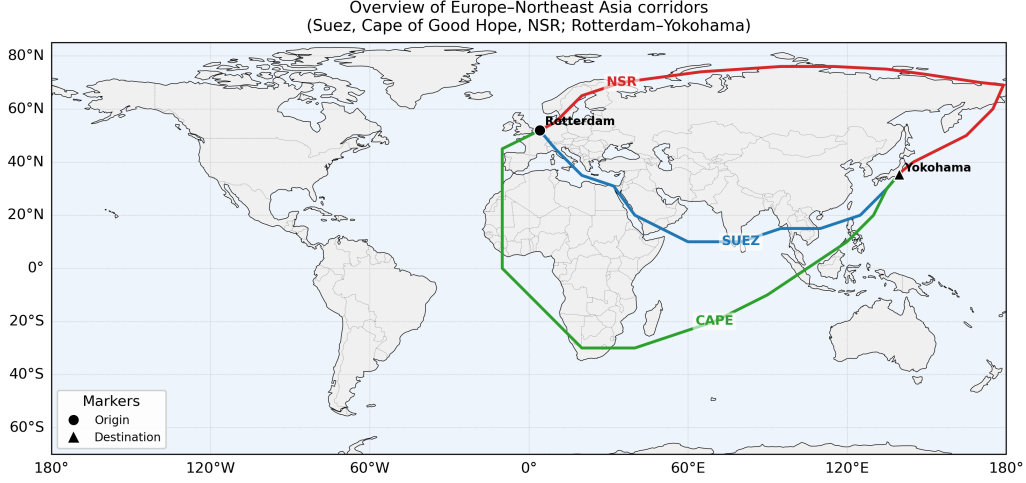


Figure 1: Overview of the three Europe–Northeast Asia corridors considered in this study

This ensures both operational relevance and comparability to earlier work, while magnifying routing-method effects over intercontinental distance.

We explicitly include three strategic maritime corridors:

- (i) **Suez Canal Corridor (SUEZ)**: the established Europe–Asia trunk route via the Mediterranean, Suez Canal, Red Sea, and Indian Ocean;
- (ii) **Cape of Good Hope Corridor (CAPE)**: a canal-independent fall-back via the Atlantic and Southern Indian Oceans, used during Suez and Red Sea disruptions;
- (iii) **Northern Sea Route (NSR)**: the high-latitude Arctic corridor along the Russian coast between the Barents Sea and Bering Strait.

A global overview of these three corridors for the Rotterdam–Yokohama origin–destination pair is shown in Figure 1.

For context, the direct great-circle distance between Rotterdam and Yokohama is about 5.1 k nm, i.e. substantially shorter than any of the corridorised routes considered here.

We do not include the Panama Canal or the Northwest Passage. For the selected Northern Europe–Northeast Asia trade, Panama is not a competitive option, and the Northwest Passage remains operationally rare and heavily constrained. Including them would add complexity without materially changing the main corridor competition for this trade.

Table 1: Macro waypoint manifest and routing-variant philosophies for each corridor. Variants correspond to (a) service-guided, (b) bluewater-oriented, and (c) channel/coast-guided waypoint chains used to define corridor-specific GC and sea-only routes.

scenario	corridor	philosophy	n_waypoints	waypoints_chain
CAPE_VAR1_service	CAPE	VAR1	10	(52.0,4.1) → (36.0,-5.6) → (14.7,-17.5) → (-34...
CAPE_VAR2_bluewater	CAPE	VAR2	9	(52.0,4.1) → (30.0,-20.0) → (-10.0,-25.0) → (-3...
CAPE_VAR3_coast	CAPE	VAR3	12	(52.0,4.1) → (36.0,-5.6) → (5.0,-5.0) → (-20.0,...
NSR_VAR1_service	NSR	VAR1	9	(52.0,4.1) → (71.0,25.0) → (70.5,58.0) → (76.0,...
NSR_VAR2_bluewater	NSR	VAR2	9	(52.0,4.1) → (73.0,35.0) → (70.5,58.0) → (76.5,...
NSR_VAR3_coast	NSR	VAR3	9	(52.0,4.1) → (70.0,33.0) → (70.5,58.0) → (75.0,...
SUEZ_VAR1_service	SUEZ	VAR1	10	(52.0,4.1) → (50.5,1.0) → (36.0,-5.6) → (30.0,3...
SUEZ_VAR2_bluewater	SUEZ	VAR2	10	(52.0,4.1) → (48.0,-10.0) → (36.0,-5.6) → (30.0...
SUEZ_VAR3_coast	SUEZ	VAR3	13	(52.0,4.1) → (50.5,1.0) → (36.0,-5.6) → (33.5,2...

For each corridor we define three operational *waypoint philosophies*:

- (a) a *service-guided* variant that loosely follows typical liner schedules and gateway regions (major hubs and chokepoints);
- (b) a *bluewater-oriented* variant favouring longer open-ocean legs where feasible; and
- (c) a *channel/coast-guided* variant that adheres more closely to constrained straits and coastal guidance.

These nine corridor-variant combinations bracket plausible operator routing styles without requiring detailed commercial schedules. Waypoints were defined manually based on contemporary sailing directions and publicly available track patterns, then verified via visual inspection on interactive charts to ensure they represent realistic chokepoints and gateway regions rather than arbitrary mid-ocean points. They allow us to assess whether the effects of routing representation (geometric vs. sea-only) persist across reasonable corridor geometries. The Macro waypoints and variant philosophies are summarized in Table 1.

2.2. Routing representations: geometric GC vs. sea-only A^*

We compare two routing representations for each corridor and waypoint variant.

2.2.1. Great-circle baselines

Great-circle (GC) paths represent the geometric minimum distance between two points on a sphere. They are widely used in maritime and transport studies as distance benchmarks and as inputs to cost and emission

models. However, raw GC arcs do not incorporate land–sea geometry, do not respect coastlines or chokepoints, and in principle may cross continental landmasses. In practice, GC-based corridor distances often reflect either (i) a single GC between origin and destination or (ii) a sum of GCs between a small set of corridor waypoints. In both cases, GC distances act as *optimistic lower bounds* that idealise the Earth as a smooth sphere and abstract away navigational feasibility.

In this study, we construct for each corridor and waypoint variant a *GC chain* connecting Rotterdam, the corridor-specific macro-waypoints, and Yokohama in sequence. These chains capture the geometric potential of each corridor and are representative of the kind of distances often invoked when the NSR is described as “30–40% shorter than Suez”.

2.2.2. Sea-only routing with A^*

To obtain physically feasible paths, we use the open-source `searoute` package (Halili, 2022), which implements a Dijkstra/ A^* shortest-path search on a global oceanic graph. Internally, `searoute` uses a regular latitude–longitude grid (approximately 0.5° resolution) whose nodes are constrained to water cells; coastlines and land are masked out when constructing the graph (as implemented in `searoute`’s default coastline mask). Edges connect neighbouring water nodes with weights proportional to great-circle segment length.

For each consecutive waypoint pair in a corridor’s macro-waypoint chain, `searoute` is used to compute a sea-only shortest path. The resulting segments are concatenated to form a continuous sea-only polyline from Rotterdam to Yokohama. Every step of the search is constrained to water nodes; land and narrow straits act as hard constraints. This sea-only representation therefore captures corridor geometry, including high-latitude bends and mandatory chokepoints along each route (e.g. Bering Strait on the NSR, the Suez Canal, and the Cape of Good Hope), and provides an operationally meaningful approximation of feasible vessel trajectories. We verified algorithmic parity by confirming that A^* and Dijkstra return identical shortest-path distances on the same sea graph.

The effective grid resolution ($\approx 0.5^\circ$) is chosen as a compromise between computational cost and geometric fidelity. At intercontinental scales, corridor-level differences in distance and time are insensitive to finer resolutions, whereas substantially coarser grids risk misrepresenting narrow passages or coastal stand-off. At 60°N , a 0.5° cell corresponds to roughly 15–

30 nm spacing, adequate for corridor-scale benchmarking. Because our aim is to isolate *routing-method effects* rather than resolve harbour approaches or traffic separation schemes (TSS), this resolution is sufficient.

Antimeridian crossings (near $\pm 180^\circ$ longitude) are handled through a combination of `searoute`’s internal graph representation and explicit date-line guards in post-processing, ensuring that polylines are continuous and free of wrap-around artefacts. Route geometries were validated by segment-length diagnostics to confirm no wrap-around jumps. Route polylines are subsequently interpolated to a uniform spacing (0.25°) for consistent distance estimation.

2.2.3. Rationale for comparing GC and A*

Comparing GC and A* isolates the effect of *routing representation* itself: to what extent widely reported corridor advantages (e.g. “NSR is 30–40% shorter than Suez”) derive from geometric idealisations rather than navigable paths. GC chains bound the problem from below as idealised geometric minima. Sea-only A* routes represent the shortest navigable paths under static constraints, providing a conservative baseline prior to adding ice or weather costs.

Alternative graph-search or optimisation methods (e.g. Dijkstra on the same graph, dynamic programming, evolutionary heuristics, reinforcement learning) would return the same shortest path as A* under identical static constraints on a single-objective distance cost. Their added value lies in multi-objective or metocean-aware optimisation, which we deliberately reserve for subsequent work on dynamic voyage planning and environmental routing. The goal here is to cleanly separate the effect of moving from geometric GC representations to sea-only, coastline-respecting paths and to quantify how this transition propagates into distance, time, fuel, and CO₂ across the three corridors.

2.3. From distance to time, fuel, and CO₂

For each scenario (corridor $\times$ waypoint philosophy $\times$ routing method), total route distance D [nm] is computed as the sum of great-circle segment lengths along the GC chain or A* sea-only polyline.

Indicative voyage time is then obtained using corridor-specific average service speeds U_{corridor} representative of current practice:

$$T = \frac{D}{U_{\text{corridor}}}, \quad (1)$$

where T is the sailing time. We do not optimise speeds; instead, we fix representative values to highlight how routing-method differences interact with realistic speed policies:

- NSR: 12.5 kn (reflecting ice-class and high-latitude operational limits) (Shu et al., 2024; Li et al., 2024);
- SUEZ: 14.5 kn (typical mainline container service under slow-steaming practice) (BIMCO, 2024);
- CAPE: 14.0 kn (open-ocean service; conservative relative to Cape-rerouting averages) (BIMCO, 2024).

These values reflect corridor-typical slow-steaming practice and Arctic speed limits and are used to isolate routing-method effects rather than optimise operations (BIMCO, 2024; Shu et al., 2024).

Fuel consumption and CO₂ emissions are derived from voyage time using explicit main- and auxiliary-engine accounting. For the main engine, we assume a cubic speed–power relation and scale power from a reference operating point (P_0, U_0) :

$$P_{\text{ME}} = P_0 \left(\frac{U}{U_0} \right)^3, \quad \dot{m}_{\text{ME}} = P_{\text{ME}} \cdot \text{SFOC}_{\text{ME}}, \quad (2)$$

where (P_0, U_0) is a reference condition (Table 2). SFOC values are converted from g kWh^{−1} to t h^{−1} using standard unit factors. Total main-engine fuel is $m_{\text{ME}} = \dot{m}_{\text{ME}} T$. Auxiliary-engine fuel is modelled as a constant power load over time, $m_{\text{AUX}} = \dot{m}_{\text{AUX}} T$, capturing hotel and support loads during the voyage. The combined fuel consumption is then

$$m_{\text{FUEL}} = m_{\text{ME}} + m_{\text{AUX}}. \quad (3)$$

CO₂ emissions are computed using a standard emission factor EF_{CO_2} [t CO₂/t fuel]:

$$m_{\text{CO}_2} = m_{\text{FUEL}} \cdot EF_{\text{CO}_2}. \quad (4)$$

The cubic speed–power relation, constant SFOC, and CO₂ emission factor follow common practice in shipping emission inventories and IMO GHG studies (European Environment Agency, 2019; International Maritime Organization, 2020; Smith et al., 2014). This level of detail is sufficient for iso-

Table 2: Assumed vessel, engine, and fuel parameters for the indicative time–fuel–CO₂ calculations (Panamax / small post-Panamax container vessel). Values can be adapted to other ship types without altering the methodology.

Symbol	Description	Value (example)
P_0	Reference main-engine power	35 000 kW
U_0	Reference service speed	14.5 kn
$SFOC_{ME}$	Main engine SFOC	170 g kWh ⁻¹
P_{AE}	Auxiliary power	2 000 kW
$SFOC_{AE}$	Auxiliary SFOC	185 g kWh ⁻¹
EF_{CO_2}	CO ₂ factor (HFO)	3.114 t CO ₂ /t fuel
U_{NSR}	Representative NSR speed	12.5 kn
U_{SUEZ}	Representative Suez speed	14.5 kn
U_{CAPE}	Representative Cape speed	14.0 kn

lating routing-method effects on voyage time, fuel, and CO₂; higher-fidelity resistance and engine models are reserved for subsequent work.

We intentionally retain a simple, transparent fuel and emission model so that the influence of routing representation on T , m_{FUEL} , and m_{CO_2} is fully traceable. Ship-specific resistance, weather-dependent added resistance, and speed-control strategies are left for follow-up work.

Representative parameter values (for a Panamax container vessel of approximately 50 000–70 000 DWT on conventional HFO) are summarised in Table 2 and are consistent with standard practice in shipping emission inventories and IMO guideline assumptions (e.g. International Maritime Organization, 2020; European Environment Agency, 2019; Smith et al., 2014).

A Panamax-class container vessel is used as a neutral baseline to convert voyage time into indicative fuel and CO₂. Absolute totals depend on ship size; however, under first-order scaling with common speed–power and SFOC assumptions, corridor rankings and relative routing-method deltas are expected to remain stable. Vessel-size sensitivity is examined in follow-on work. The parameter values are chosen to represent a generic Panamax vessel rather than a specific ship, so results reflect typical corridor-level behaviour instead of vessel-specific performance. The exact numerical values can be updated without changing the structure of the workflow.

All computations were performed in Python 3.11 using `pandas`, `geopy`, `shapely`, and `searoute`. Derived data tables were exported as CSV and

L^AT_EX tables, and visualised through static charts and interactive `folium` maps.

2.4. Scenario matrix and sensitivity tests

The primary scenario matrix consists of:

- three corridors (SUEZ, CAPE, NSR);
- three waypoint philosophies per corridor (service-guided, bluewater-oriented, channel/coast-guided);
- two routing representations (GC chain and A* sea-only); and
- corridor-specific average service speeds as in Table 2.

For each element in this matrix, we compute distance, indicative time, fuel, and CO₂ as described above.

To assess robustness, we perform several complementary sensitivity analyses:

- Endpoint sensitivity:** repeating the analysis with an East Asia endpoint in Busan instead of Yokohama to assess how NSR’s relative advantage changes for more northerly/westerly destinations.
- Equal-speed comparisons:** repeating the time and fuel calculations with a common service speed (14 kn) across all corridors to isolate geometric and routing effects from speed policy.

We deliberately exclude dynamic environmental and regulatory constraints (sea ice, winds, waves, currents, emission control areas, canal queues, ice pilotage/escort requirements) to maintain focus on routing-method effects. These factors will be layered onto the A* sea-only baseline in subsequent work on dynamic Arctic voyage planning.

2.5. Computational workflow overview

Figure 2 summarises the computational pipeline. Predefined macro-waypoints feed into GC and sea-only (A*) routing, from which distance–time–fuel–CO₂ metrics are derived under corridor-specific speeds. All intermediate outputs (routes, metrics, tables, and maps) are exported for reproducibility and further analysis.

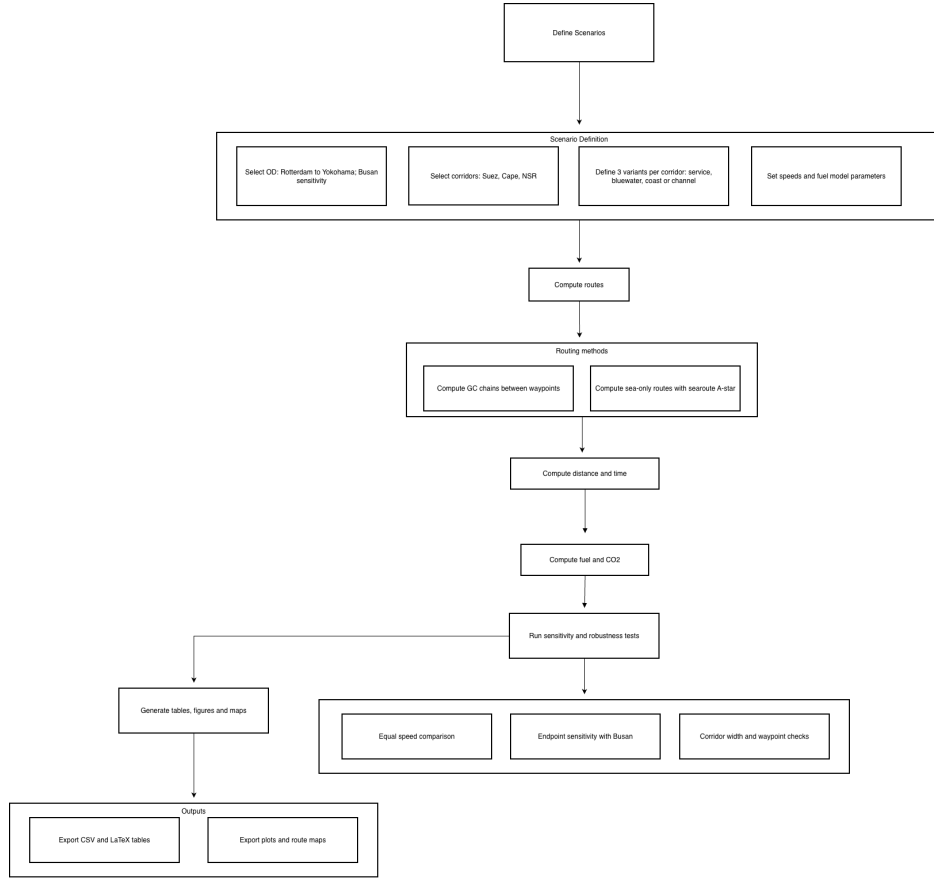


Figure 2: Overview of the methodological pipeline used to compute GC and sea-only routes, derive time/fuel/CO₂ metrics, and generate reproducible tables and maps.

3. Results and discussion

3.1. Case 1: Distance baselines and routing-method effects

Enforcing sea-only feasibility preserves the qualitative distance ranking **Northern Sea Route** < **Suez** < **Cape of Good Hope** across all waypoint variants as shown in Figure 3 and Figure 4 with per variant metrics in (Table 3) and corridor-level medians in (Table 4). For Suez and Cape, GC-chain and sea-only distances remain close, reflecting mostly open-ocean legs. For the Northern Sea Route, the sea-only constraint introduces a larger adjustment because routing must respect Arctic coastlines, straits, and the Bering gateway.

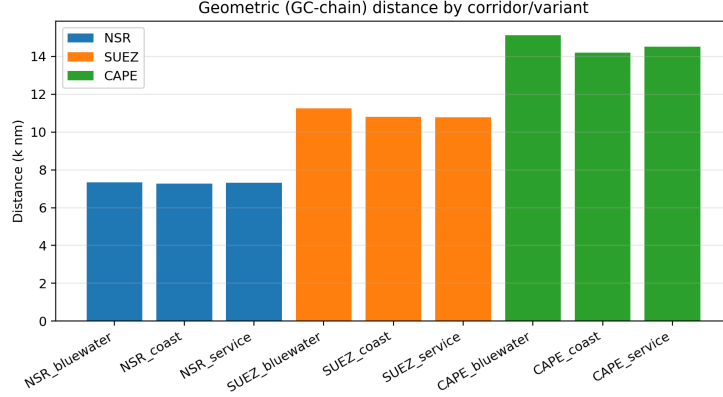


Figure 3: Geometric GC-chain corridor distances by corridor and waypoint variant for Rotterdam–Yokohama.

GC-chain to sea-only deltas are modest for Suez ($\sim 6.4\text{--}8.4\%$) and Cape ($\sim 6.2\text{--}13.2\%$), but substantially larger for the Northern Sea Route ($\sim 17.4\text{--}18.2\%$) (Figure 6). These deltas confirm that geometry-only baselines materially underestimate navigable Northern Sea Route length, while having a smaller influence on the lower-latitude corridors.

Sea-only distances (Figure 4) and their medians (Table 4) confirm that the Northern Sea Route remains shortest by nautical miles. However, for this Rotterdam–Yokohama pair the headline geometric advantage quoted in the literature (often $\sim 30\text{--}40\%$ from pure geometry) *shrinks materially* once sea-only feasibility and realistic corridor geometries are enforced, yielding an effective advantage of roughly $\sim 25\text{--}30\%$ versus Suez depending on variant.

Within-corridor spread across the three waypoint philosophies is small (Figures 3 and 4); the distance coefficient of variation is only a few percent (median CV around 2–3%), indicating that the main findings are not an artefact of a single waypoint choice.

3.2. Case 2: Downstream propagation to time, fuel, and CO_2 under corridor-typical speeds

Case 1 showed that enforcing sea-only feasibility preserves the distance ordering ($\text{NSR} < \text{SUEZ} < \text{CAPE}$), while revealing non-trivial GC $\rightarrow$ sea-only corrections. Case 2 addresses the paper’s secondary objective by tracing how these sea-only distance baselines propagate into indicative voyage time and the resulting fuel and CO_2 estimates under corridor-typical operating speeds.

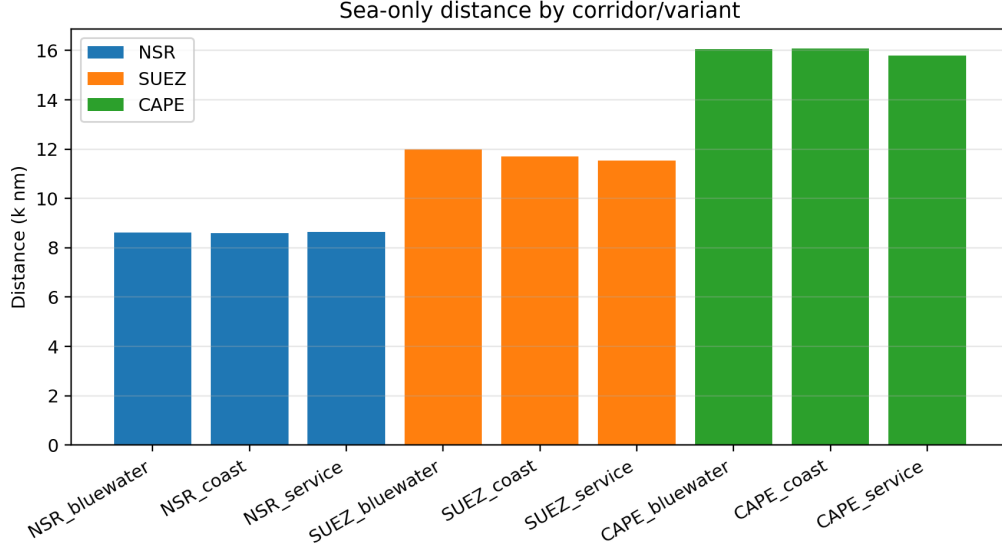


Figure 4: Sea-only (A^*) distance by corridor and waypoint variant. The Northern Sea Route remains shortest, followed by Suez and Cape.

Sea-only distances are converted to days at sea using representative corridor speeds (NSR ≈ 12.5 kn; SUEZ ≈ 14.5 kn; CAPE ≈ 14 kn; Table 2). Figure 7 shows that distance does not translate linearly into schedule outcomes once realistic speed policies are applied. Although NSR remains shortest by nautical miles, its lower corridor-typical speed compresses the schedule advantage relative to SUEZ. Across variants, indicative durations cluster around ~ 29 days for NSR, ~ 33 – 35 days for SUEZ, and ~ 47 – 48 days for CAPE, with only modest within-corridor spread. Thus, a nominal 25–30% NSR distance advantage corresponds to only a 15% time advantage in this fair-weather baseline.

The compression is made explicit in Figure 8, which compares corridor medians normalised by SUEZ. NSR’s median sea-only distance is about $0.74\times$ SUEZ, but the corresponding time is only $0.85\times$ SUEZ. Conversely, CAPE is about $1.37\times$ SUEZ in distance and $1.42\times$ in time. This directly supports the core message of the paper: *shorter distance is an optimistic proxy for operational advantage unless speed policy is stated explicitly*.

Figure 9 provides a complementary view: if all corridors sailed at SUEZ speed (dashed reference line), NSR points would fall on the same distance–time trend, but under NSR’s lower speed they sit above that line. This visualises

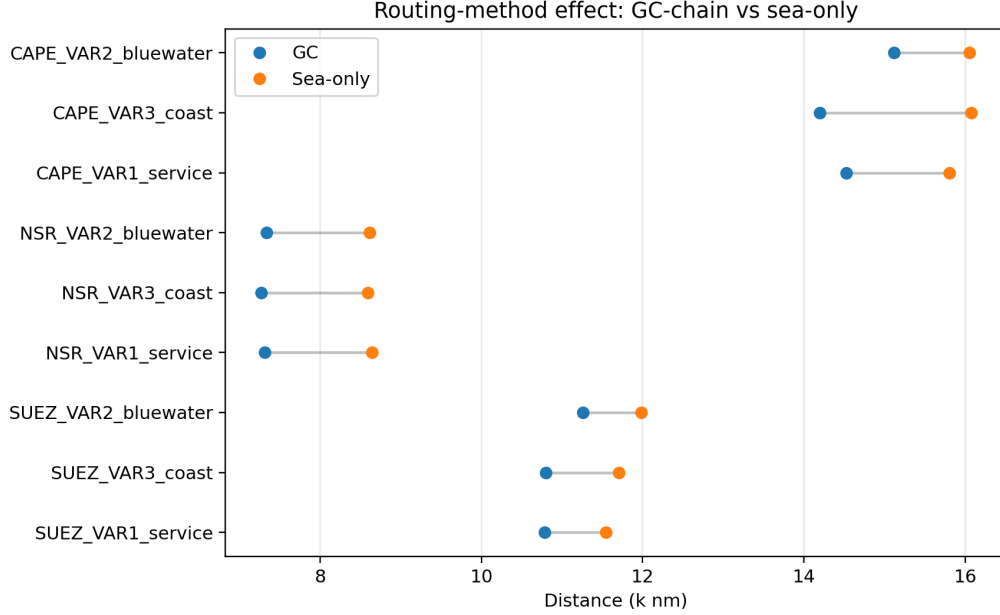


Figure 5: Routing-method effect on distance: GC-chain vs sea-only per corridor/variant.

the non-linear distance–time mapping that motivates the remainder of the analysis.

To quantify the speed threshold behind this compression, the break-even analysis in Figure 10 shows the NSR service speed required to match SUEZ duration for each paired variant. The required NSR speed is ~ 10.4 – 10.9 kn (median ~ 10.7 kn). Thus, for Rotterdam–Yokohama the NSR remains schedule-advantaged at the assumed 12.5 kn, but *only modestly*; a reduction of NSR average speed below the break-even band would remove the time advantage entirely.

Fuel and emissions are computed from voyage hours using the transparent main- plus auxiliary-engine accounting in Section 2.3. Figure 11 shows that total fuel (and thus CO_2) follows time rather than distance alone. Absolute totals preserve the ordering $\text{NSR} < \text{SUEZ} < \text{CAPE}$, but the NSR margin over SUEZ is smaller than the distance gap because corridor-typical speeds reduce the hours-at-sea advantage. Indicative totals are approximately ~ 0.92 kt for NSR, ~ 1.9 kt for SUEZ, and ~ 2.4 kt for CAPE, implying roughly ~ 2.9 , 5.9 , and 7.5 kt CO_2 using the standard factor. Within-corridor variation across waypoint philosophies remains small, showing that the downstream

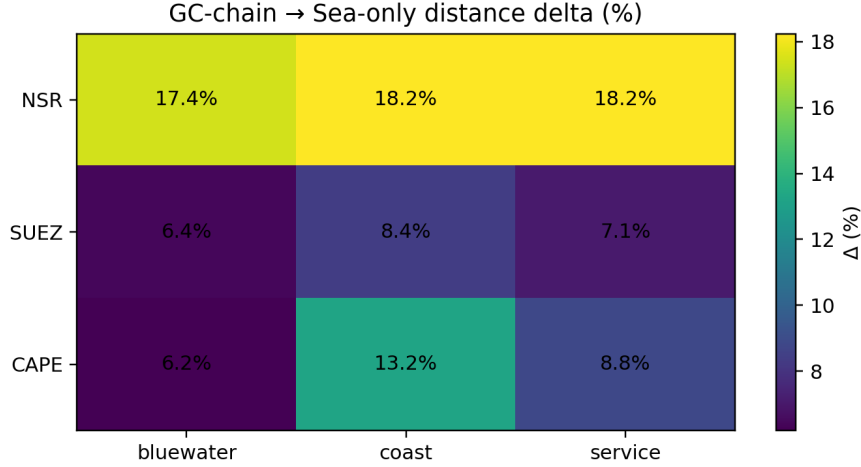


Figure 6: GC-chain to sea-only distance delta (%) by corridor and waypoint philosophy.

Table 3: Baseline metrics per variant: GC-chain vs sea-only distance, and derived indicative time and fuel/CO₂.

corridor	Variant	variant	GC (k nm)	Sea-only (k nm)	Δ (nm)	GC→Sea Δ (%)
CAPE	CAPE_VAR1_service	service	14.52	15.80	1278.70	8.80
CAPE	CAPE_VAR2_bluewater	bluewater	15.11	16.05	937.61	6.20
CAPE	CAPE_VAR3_coast	coast	14.20	16.07	1876.24	13.20
NSR	NSR_VAR1_service	service	7.31	8.64	1333.23	18.20
NSR	NSR_VAR2_bluewater	bluewater	7.34	8.61	1276.45	17.40
NSR	NSR_VAR3_coast	coast	7.27	8.59	1323.14	18.20
SUEZ	SUEZ_VAR1_service	service	10.78	11.55	764.01	7.10
SUEZ	SUEZ_VAR2_bluewater	bluewater	11.26	11.98	722.94	6.40
SUEZ	SUEZ_VAR3_coast	coast	10.80	11.70	905.80	8.40

conclusions are robust to plausible corridor realisations.

Overall, Case 2 demonstrates the downstream consequence of routing-method choice: geometric distance advantages provide useful lower bounds, but once converted to operational metrics under corridor-typical speeds, the NSR advantage for this Europe–Asia origin–destination pair compresses substantially. This completes the study objective by showing how routing representation and operating policy jointly determine corridor competitiveness in time, fuel, and emissions.

Table 4: Sea-only distance medians by corridor across all waypoint variants (Rotterdam–Yokohama).

corridor	Median (k nm)	Min (k nm)	Max (k nm)	std	CV (%)
CAPE	16.05	15.80	16.07	150.27	0.90
NSR	8.61	8.59	8.64	27.20	0.30
SUEZ	11.70	11.55	11.98	220.65	1.90

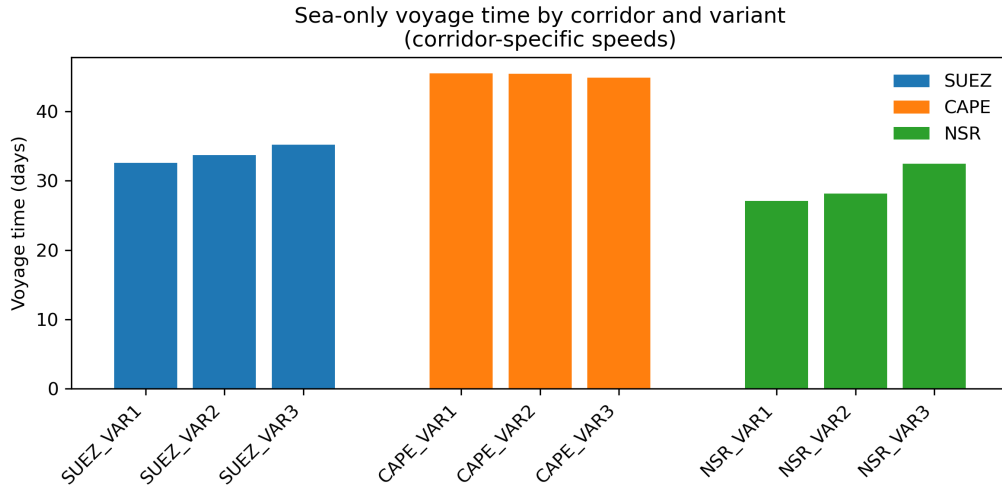


Figure 7: Voyage duration by corridor and variant for sea-only routes under corridor-typical service speeds. NSR remains shortest by distance, but its time advantage over SUEZ compresses under the lower NSR speed.

3.3. Case 3: Sensitivity and robustness

Case 3 tests whether the distance and downstream conclusions from Cases 1–2 remain stable under plausible changes in endpoint location and operating-speed policy.

Figure 12 compares sea-only distances when the East Asia endpoint shifts from Yokohama to Busan. The shift lengthens all corridors by a similar absolute amount of about 0.64 k nm, reflecting a northwestward relocation of the destination. However, the *relative* impact differs by corridor: the Northern Sea Route increases by ≈ 7.39 – 7.44% across variants, compared with ≈ 5.33 – 5.53% for Suez and ≈ 3.97 – 4.04% for Cape. This pattern is consistent with the corridor geometry: moving to Busan shortens the northwest

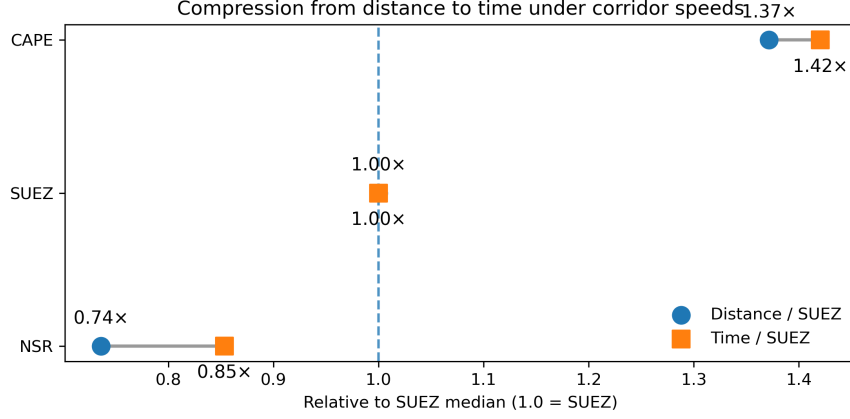


Figure 8: Compression from distance to time under corridor-typical speeds. Points show corridor medians across variants, normalised by SUEZ. NSR’s distance advantage translates into a smaller time advantage once realistic speeds are applied.

Pacific approach from the Bering Strait, so the NSR retains (and slightly enlarges) its distance advantage over Suez for more northerly/westerly East Asian markets. Importantly, the qualitative ordering is unchanged ($\text{NSR} < \text{SUEZ} < \text{CAPE}$), showing that the main distance ranking is not an artefact of a single destination choice.

To isolate pure geometry from operating policy, we recomputed indicative voyage times using a common service speed across corridors. As shown in Figure 13, equal-speed assumptions restore an almost linear distance-to-time mapping and recover the geometric ordering from Case 1. The comparison makes explicit that the compression of NSR’s schedule and fuel advantages in Case 2 is driven by corridor-typical speed differences rather than by instability in the routing baselines.

Overall, Case 3 confirms that the corridor rankings and the key downstream interpretations are robust to reasonable endpoint relocation and to the choice between corridor-typical and equal-speed operating assumptions, directly supporting the paper’s contribution of isolating routing-method effects from operational and geographic confounders.

Additional tests with perturbed corridor widths and macro-waypoint locations (not shown) changed absolute sea-only distances by at most a few percent and did not affect the $\text{NSR} < \text{SUEZ} < \text{CAPE}$ ranking or the main time–fuel– CO_2 patterns.

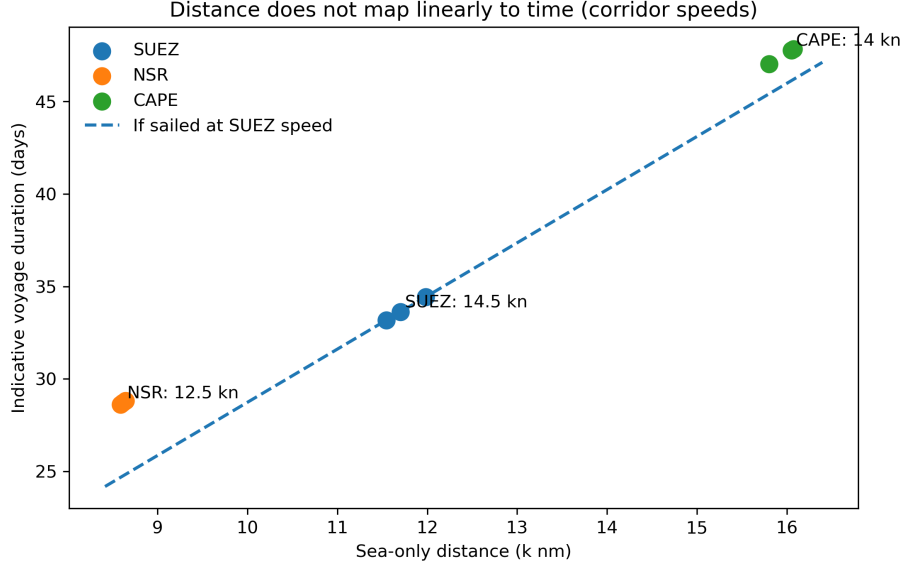


Figure 9: Sea-only distance vs. indicative voyage duration under corridor speeds. The dashed line shows the duration each route would have at SUEZ speed. NSR points lie above this line because NSR speed is lower.

3.4. Implications and limitations

For corridor selection and fleet planning, three points stand out: (i) *Corridor choice and speed policy* matter as much as raw distance; distance-based comparisons alone can materially overstate the NSR’s practical schedule and fuel advantage over SUEZ for the Rotterdam–Yokohama trade. (ii) Equal-speed comparisons are useful for isolating geometry, but *corridor-typical speeds* determine realised schedules, fuel use, and auxiliary exposure. (iii) The CAPE corridor, although longest, remains a robust year-round fallback that is independent of canals and Arctic seasonality. These implications are stable under plausible endpoint relocation (Busan vs. Yokohama) and under equal-speed sensitivity tests (Case 3).

By construction, the present analysis *excludes* dynamic environmental and regulatory constraints (sea ice, winds, waves, currents, emission-control areas, canal queues, ice-pilotage and escort requirements). These are essential for operational decision-making but are deliberately omitted here to isolate routing-method effects. The fuel and CO₂ model is transparent but not ship-specific: it does not include detailed resistance models, speed–power–SFOC

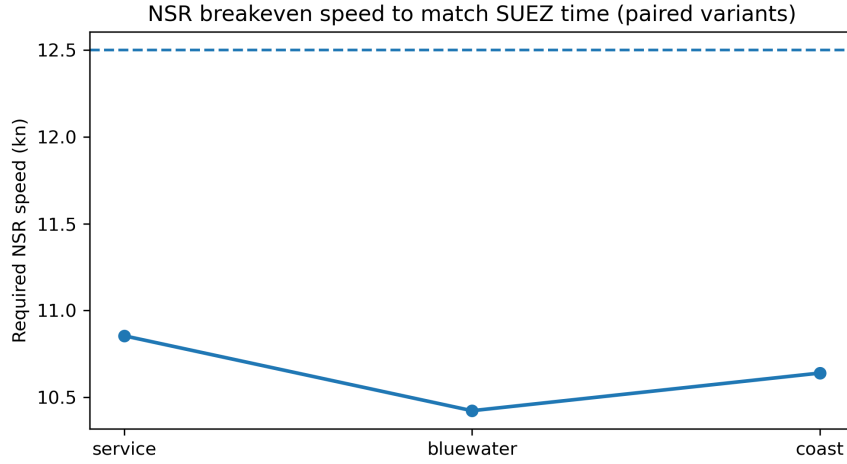


Figure 10: NSR break-even speed required to match SUEZ voyage time for paired variants. If NSR average speed drops below this band, its schedule advantage disappears.

coupling, or weather-dependent added resistance. Finally, the results are computed for a single origin–destination pair with curated waypoints; the framework is general, but quantitative deltas will vary for other trades and waypoint sets.

Overall, the results show that routing representation is a first-order driver of corridor conclusions. Geometry-only reasoning reproduces the familiar “Arctic shortcut” narrative, whereas conservative sea-only routing combined with corridor-typical speeds reveals a much more compressed and corridor-dependent picture of time, fuel, and emissions.

These fair-weather baselines provide the routing layer for subsequent work in which sea ice, metocean forcing, and regulatory overlays are introduced on top of the sea-only A* framework.

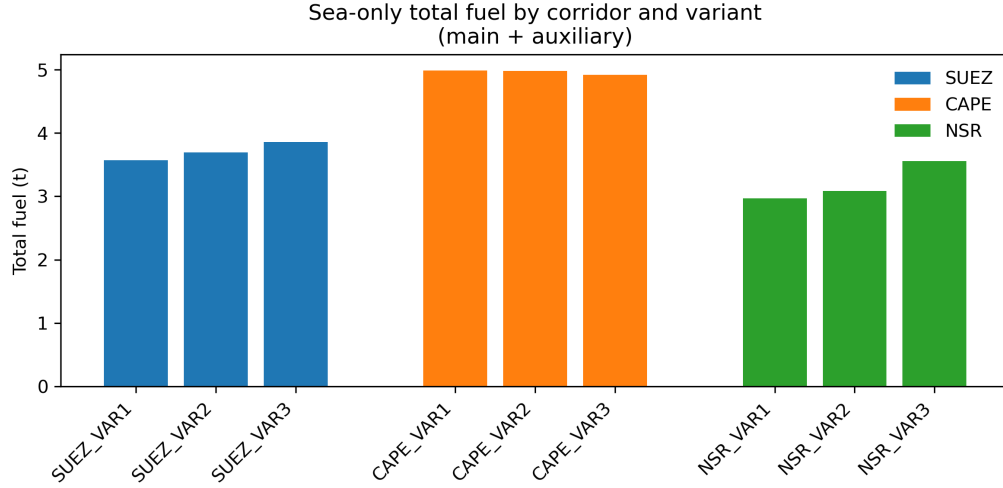


Figure 11: Estimated total fuel consumption by corridor and variant for sea-only routes (main + auxiliary). CO₂ follows the same pattern because it scales with total fuel.

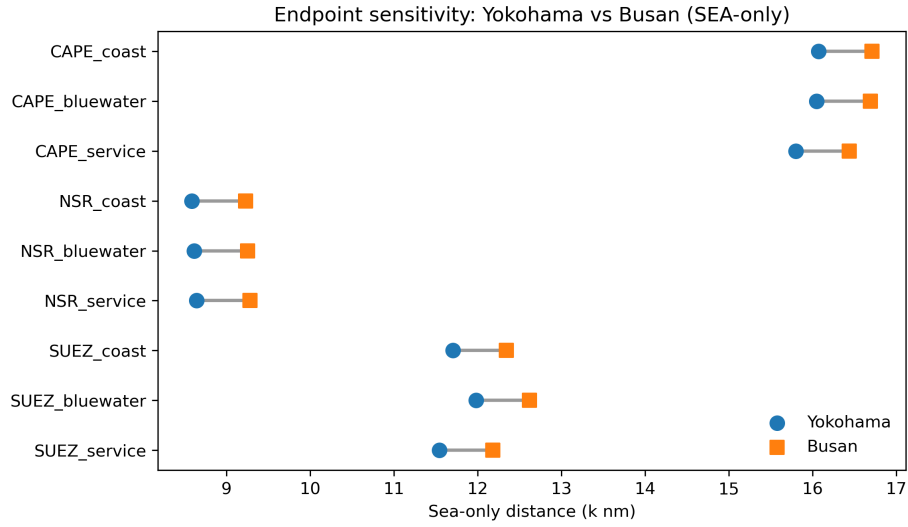


Figure 12: Endpoint sensitivity for sea-only distances: Yokohama versus Busan by corridor and waypoint variant. All corridors lengthen by ≈ 0.64 k nm, with the largest relative increase for NSR.

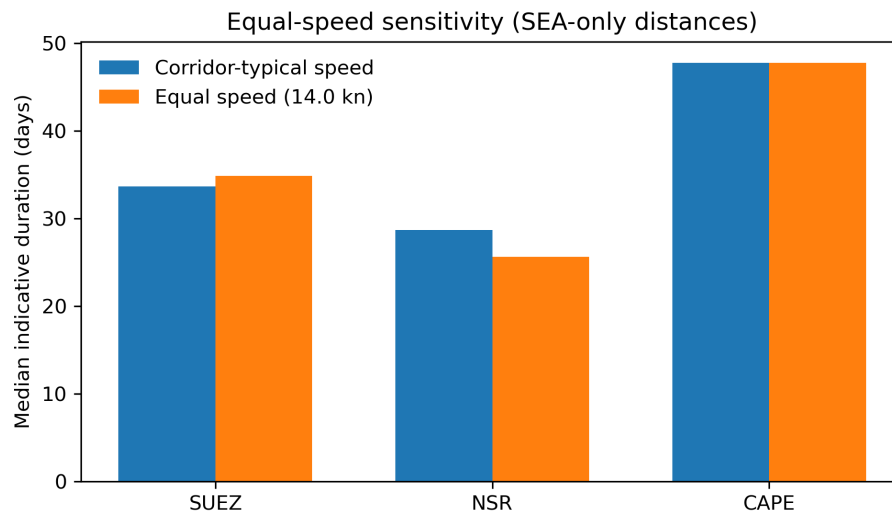


Figure 13: Equal-speed sensitivity for sea-only routes. Median durations under corridor-typical speeds are compared to an equal-speed scenario, isolating operating-policy effects from geometry.

4. Conclusions

This paper quantified how routing representation—geometric great-circle (GC) versus physically feasible sea-only paths—affects distance, time, fuel, and CO₂ estimates for Europe–Asia trade across three strategic corridors: the Suez Canal, the Cape of Good Hope, and the Arctic Northern Sea Route (NSR). A coastline-masked A* framework was applied to Rotterdam–Yokohama with three operational waypoint philosophies per corridor and a transparent main- and auxiliary-engine accounting.

Three main conclusions emerge:

1. **Sea-only routing preserves qualitative distance ordering but alters magnitude.** The ranking $\text{NSR} < \text{SUEZ} < \text{CAPE}$ holds once navigability is enforced, with small GC→sea-only adjustments for SUEZ/-CAPE and larger adjustments for NSR driven by Arctic coastal geometry and chokepoints. As a result, NSR’s widely cited geometric advantage shrinks for this OD pair under feasible routing.
2. **Distance does not translate linearly into time, fuel, or CO₂.** Under corridor-typical speeds, NSR’s schedule and fuel advantages compress and may disappear relative to SUEZ. CAPE remains longest in both distance and duration. Fuel and CO₂ primarily follow hours at sea, underscoring that distance alone is an insufficient proxy for operational or environmental performance.
3. **Competitiveness depends on endpoint and speed policy.** NSR’s relative advantage increases for more northerly East Asian ports (e.g., Busan) and decreases for Yokohama. Equal-speed comparisons recover pure geometric differences, while corridor-typical speeds show how operating practices can offset or amplify them. CAPE emerges as a robust canal- independent fallback rather than a competitive primary option.

Methodologically, the study provides a reproducible, corridor-agnostic baseline linking routing-method choice to distance, time, fuel, and CO₂ for Europe–Asia services. The findings are robust across waypoint philosophies and sensitivity tests, and the framework is designed for extension. Future work will integrate seasonal and daily sea-ice fields, metocean routing, emission-control areas and canal constraints, and ship-specific propulsion models, extending the static fair-weather baseline toward dynamic voyage-planning and digital-twin applications for Arctic and global shipping.

CRediT authorship contribution statement

Abdella Mohamed: Conceptualization, Methodology, Coding, Data curation, Writing, Visualization, Investigation, Validation. **Xiangyu Hu:** Supervision, Conceptualization, Methodology, Manuscript review and editing. **Christian Hendricks:** Mentorship, Conceptualization, Methodology, Manuscript review and editing.

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Code and data availability

All scripts used to compute distances, energy use, and to generate figures and tables are available in the project repository at [DOI/URL].

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

The author used OpenAI ChatGPT to assist with language editing during manuscript preparation. The author reviewed and edited the content and accepts full responsibility for the final manuscript.

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